



Test Of Mathematics For University Admission TMUA

INFORMATION

This is a custom TMUA paper that I (James Chen Year 12 Reading School) have made.

Typically, paper 1 assesses mathematical techniques and paper 2 assesses more on logic and proof, but this paper will have a mix of both.

The solutions are given at the bottom of the document.

Reminders:

This paper consists of 20 questions.

Each question is worth one mark and there are no penalties for incorrect answers.

You should attempt all 20 questions.

The number of options varies by question.

Calculators are not permitted.

There is no formulae booklet.

Time Allowed: 75 Minutes

1.

What is the gradient of the perpendicular bisector of the line segment connecting the points $(-3, 5)$ and $(-1, 9)$?

A -0.5

B 0.5

C -2

D 2

2.

How many distinct real solutions does the equation $x^8 = 2x^4 + 4$ have?

A 0

B 1

C 2

D 4

E 6

F 8

3.

If $\pi^2 \approx g$, $\pi > 0$, then which estimate for $\ln(\pi)$ can be deduced?

A $\ln(\pi) \approx \frac{g}{2}$

B $\ln(\pi) \approx \ln(\sqrt{g})$

C $\ln(\pi) \approx \ln(g)$

D $\ln(\pi) \approx \sqrt{\ln g}$

4.

A circle in the x - y plane has radius r , perimeter P , and area A . Which of these statements is true?

- A $r > 2$ is a necessary but not sufficient condition for $P < A$
- B $r > 0$ is a necessary and sufficient condition for $P < A$
- C $0 < r < 2$ is a sufficient but not necessary condition for $P > A$
- D $r > 2$ is a necessary and sufficient condition for $P > A$
- E $0 < r < 2$ is a necessary and sufficient condition for $P < A$
- F $r > 2$ is a sufficient but not necessary condition for $P < A$
- G $0 < r < 2$ is a necessary and sufficient condition for $P > A$

5.

Let $I(a) = \int_0^a 2x - 3dx$. For what values of a are $I(a)$ equal to -2 ?

- A $a = 1, a = 2$
- B $a = 0, a = 2$
- C $a = 0, a = 1$
- D $a = -2, a = -1$
- E $a = -1, a = 2$
- F $a = 0.5, a = 1$
- G $a = -0.5, a = 1$
- H $a = -2, a = 1$

6.

According to One Direction, you are beautiful iff you do not know you are beautiful. Also, if you flip your hair, One Direction will feel overwhelmed. Which of these statements must be true?

- A** If One Direction has felt overwhelmed, you have flipped your hair.
- B** If you do not know you are beautiful, it is possible that you are not beautiful.
- C** It is not possible for you to be beautiful.
- D** If One Direction has never felt overwhelmed, you have not flipped your hair.
- E** If you have flipped your hair or you do not know that you are beautiful, you are beautiful and One Direction has felt overwhelmed.
- F** It is not possible for you to have overwhelmed One Direction, but not be beautiful.

7.

What is the limit of $\frac{(x^2+5x+6)(x-1)}{(x+3)(x^2-1)}$ as $x \rightarrow \infty$?

- A** 0
- B** 1
- C** -1
- D** 2
- E** -2
- F** ∞
- G** $-\infty$
- H** Undefined

8.

The line $y = x$ is tangent to the curve $y = ax^2 + 5x + 1$ at the point with what x-coordinate?

A -0.5

B 5

C -1

D 1

E -2

F 2

9.

Evaluate $\sum_{r=0}^{\infty} \left(\frac{3}{10^r} + \frac{1}{10^{r+1}} + \frac{4}{10^{r+2}} \right)$.

A 3.14

B $\frac{3.14}{0.9}$

C 31.4

D $\frac{3.14}{9}$

E $\frac{3.14}{0.09}$

F $\frac{0.9}{3.14}$

G $\frac{3.14}{1.1}$

H $\frac{1.1}{3.14}$

10.

A coffee shop's 8 workers: Emily, Ethan, Olajide, Pranav, Rujul, Sidhant, Simon and Sabrina are all wearing T-shirts with the first of letters of their names printed on the fronts. How many different arrangements of the 8 workers causes their T-shirts to spell 'ESPRESSO' from left to right?

- A 1
- B 2
- C 4
- D 6
- E 8
- F 12
- G 16
- H 24

11.

Which function $f(x)$ satisfies the equation $\sqrt[4]{abcd} = f^{-1}\left(\frac{f(a)+f(b)+f(c)+f(d)}{4}\right)$, where $f^{-1}(x)$ is the inverse function of $f(x)$?

- A $f(x) = x$
- B $f(x) = x^2$
- C $f(x) = \log_{2025}(x)$
- D $f(x) = (\ln(x))^2$
- E $f(x) = \ln(x^2)$
- F $f(x) = e^x$

12. Which line in this false proof that $1 = 2$ contains the first mistake?

$$\text{Let } a = b \quad (1)$$

$$a^2 = ab \quad (2)$$

$$a^2 - b^2 = ab - b^2 \quad (3)$$

$$(a + b)(a - b) = b(a - b) \quad (4)$$

$$a + b = b \quad (5)$$

$$a + a = a \quad (6)$$

$$1 = 2 \quad (7)$$

A (1)

B (2)

C (3)

D (4)

E (5)

F (6)

G (7)

13.

The real numbers a, b are such that $a > b$. You are given that the function $f(x)$ is defined over positive real x and is either always increasing or always decreasing. Which of these conditions is sufficient but not necessary for $f(a) < f(b)$ to be true?

A $f'(x) < 0$

B $f(x)$ is a decreasing function

C $f(x) = \log_c x, 0 < c < 1$

D $f(x) = \log_c x, c > 1$

14.

I promise to do at least 5 STEP 1 questions, at least 3 STEP II questions, and at least 1 STEP III question next week. If I break my promise, which of the following statements must be true?

A I will have done fewer than 5 STEP I questions, fewer than 3 STEP II questions, and no STEP III questions

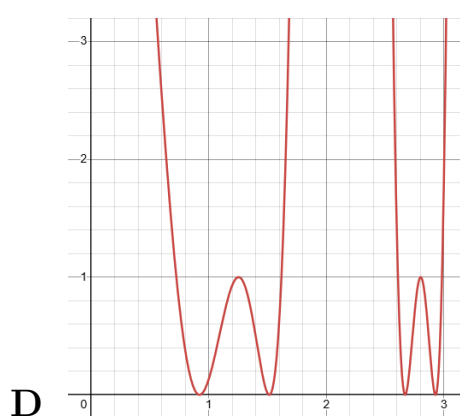
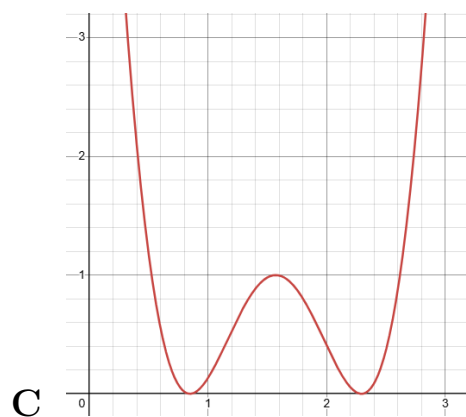
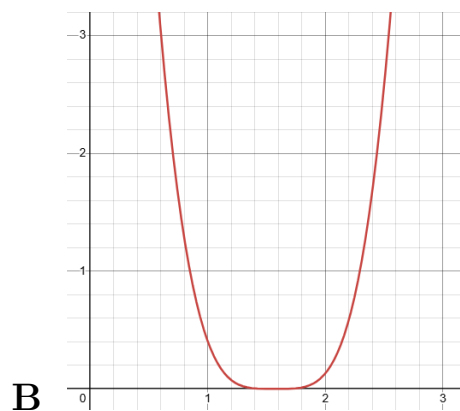
B I will have done fewer than 6 STEP I questions, fewer than 4 STEP II questions, or fewer than 2 STEP III questions

C I will have done fewer than 4 STEP I questions, fewer than 2 STEP II questions, and no STEP III questions

D I will have done fewer than 4 STEP I questions, fewer than 2 STEP II questions, or no STEP III questions

15.

Which of the following is a sketch of the graph $y = (3 - 4 \sin x)^2$?



16.

The 5 Backstreet Boys are asked if they want it that way. Their responses are as follows:

Backstreet Boy A: "Exactly one of us wants it that way."

Backstreet Boy B: "Exactly two of us wants it that way."

Backstreet Boy C: "Exactly three of us wants it that way."

Backstreet Boy D: "Exactly four of us wants it that way."

Backstreet Boy E: "Exactly five of us wants it that way."

If a Backstreet Boy does want it that way, they will tell the truth, but if a Backstreet Boy does not want it that way, they will lie. Which of these statements must be false?

A It is not necessary that at least one Backstreet Boy wants it that way.

B It is only possible that Backstreet Boy A wants it that way.

C It is not possible that two or more of the Backstreet Boys want it that way.

D It may be possible that Backstreet Boy B wants it that way.

17.

A_n and B_n are an arithmetic sequence and a geometric sequence respectively, with first terms A_1 and B_1 . You are given that $A_2 = B_3$ and $A_4 = B_2$, and that the common ratio of B_n is r . Which is a correct expression for A_1 ?

- A $\frac{B_1 r(3r-1)}{2}$
- B $\frac{B_1(3r^2-1)}{2}$
- C $\frac{B_1(1-3r^2)}{2}$
- D $\frac{B_1 r(1-3r)}{2}$
- E $\frac{B_1 r^2(3r-1)}{2}$
- F $\frac{B_1(3r-1)}{2}$
- G $\frac{B_1 r^{\frac{1}{2}}(1-3r)}{2}$
- H $\frac{B_1(1-3r)}{2}$

18.

For a given value of a , let $f(x) = x^{(x+a)^{(x+2a)^{(x+3a)^{\cdots}}}}$. What is the value of $f(0.3)$ when $a = 0.2$?

Note: $a^{b^{c^d}} = a^{(b^{(c^d)})}$, i.e. power towers are evaluated from the top to the bottom.

- A 0
- B 0.7
- C $\sqrt{0.3}$
- D ∞
- E 1
- F 0.1
- G $\sqrt{0.1}$
- H 0.3

19.

Let $f(x)$ be the only degree 5 polynomial where the coefficient of x^5 is 1 and the following equations are true:

$$f(1) = 5$$

$$f(2) = 4$$

$$f(3) = 3$$

$$f(4) = 2$$

$$f(5) = 1$$

What is the value of $f(6)$?

- A 0
- B 6
- C 30
- D 36
- E 72
- F 120

20.

How many distinct positive integer pairs, (a, b) , with $a > b$, solve the equation:

$$a^2 = b^2 + 2025?$$

Note: you are given that $2025 = 45^2 = 3^4 \times 5^2$.

- A 0
- B 1
- C 2
- D 3
- E 5
- F 7
- G 35
- H 105

Solutions

1.

The gradient of the line segment is given by $\frac{9-5}{-1-(-3)} = \frac{4}{2} = 2$ and so the gradient of any line perpendicular to it is the negative reciprocal, which is $-\frac{1}{2}$, i.e. **Option A**.

2.

We rearrange to get $x^8 - 2x^4 - 4 = 0$. Therefore, $x^4 = \frac{2 \pm \sqrt{20}}{2} = 1 \pm \sqrt{5}$. $1 + \sqrt{5} > 0$ so $x^4 = 1 + \sqrt{5}$ gives two solutions. $1 - \sqrt{5} < 0$ so $x^4 = 1 - \sqrt{5}$ gives no solutions. Therefore, the answer is **Option C**.

3.

We have $\pi^2 \approx g$ so taking the natural log of both sides give $\ln(\pi^2) \approx \ln(g)$ and using laws of logarithms, $2\ln(\pi) \approx \ln(g) \Rightarrow \ln(\pi) \approx \ln(\sqrt{g})$ which gives **Option B**. We could also obtain the answer by square rooting then taking the natural log of both sides.

4.

$P = 2\pi r$, $A = \pi r^2$ so $P > A \Rightarrow r < 2$ as $r \neq 0$. Since the area increases as the radius increases, and $r > 0$, the answer is **Option G**.

5.

$I(a) = \int_0^a 2x - 3dx = [x^2 - 3x]_0^a = a^2 - 3a$
 $a^2 - 3a + 2 = (a - 2)(a - 1) = 0 \Rightarrow a = 1, a = 2$ so the answer is **Option A**.

6.

The contrapositive of a statement is true if and only if the statement is true. Therefore, if One Direction has not felt overwhelmed, you have not flipped your hair, which gives **Option D**. Option A need not be true as other factors may overwhelm One

Direction. Option B cannot be true as the first statement given is an if and only if condition. Option C is clearly false as it is possible that you are beautiful (in which case you would not know it). Option E need not be true as only having flipped your hair or only not knowing you are beautiful does not guarantee both resulting effects are caused. Option F is false as the two statements are independent of each other (despite any semantic overlap).

7.

By factorisation and noticing the difference of two squares, $\frac{(x^2+5x+6)(x-1)}{(x+3)(x^2-1)} = \frac{x+2}{x+1}$. Intuitively, the numerator and denominator will approach equality as $x \rightarrow \infty$ and so the limit is 1 which is **Option B**. This can also be shown using L'Hôpital's Rule, which is beyond the scope of the TMUA.

8.

The gradient of $y = x$ is 1 and the gradient of $y = ax^2 + 5x + 1$ is $2ax + b$ so $2ax + 5 = 1 \Rightarrow x = \frac{-2}{a}$. There is only one point of intersection so the discriminant of $x = ax^2 + 5x + 1$ is 0, therefore $16 - 4a = 0$ so $a = 4$ and therefore $x = -0.5$, which is **Option A**.

9.

$\sum_{r=0}^{\infty} \left(\frac{3}{10^r} + \frac{1}{10^{r+1}} + \frac{4}{10^{r+2}} \right) = \sum_{r=0}^{\infty} \frac{3}{10^r} + \sum_{r=0}^{\infty} \frac{1}{10^{r+1}} + \sum_{r=0}^{\infty} \frac{4}{10^{r+2}}$. The first sum is a geometric series with first term 3 and common ratio 0.1. The second sum is a geometric series with first term 0.1 and common ratio 0.1. The third sum is a geometric series with first term 0.04 and common ratio 0.1. Therefore, the entire sum is $\frac{3}{1-0.1} + \frac{0.1}{1-0.1} + \frac{0.04}{1-0.1} = \frac{3.14}{0.9}$ which gives **Option B**.

10.

Each worker has a different name but some share a first letter. Since P, R, and O only appear once in 'ESPRESSO', Pranav, Rujul, and Olajide have fixed positions. There are two Es, and three Ss, so the number of arrangements is $2! \times 3! = 12$ which is **Option F**.

11.

We try each option one by one until we find that the answer is **Option C**.

$$f(x) = \log_{2025}(x) \Rightarrow f^{-1}(x) = 2025^x \\ 2025^{\frac{\log_{2025}(a) + \log_{2025}(b) + \log_{2025}(c) + \log_{2025}(d)}{4}} = 2025^{\frac{\log_{2025}(abcd)}{4}} = 2025^{\log_{2025}(\sqrt[4]{abcd})} = \sqrt[4]{abcd}.$$

In fact, this method works with any base, and this derives the geometric mean.

12.

We shall scrutinise each line carefully. There is nothing wrong with (1) for now as it is simply an identity we are taking to be true. Lines (2) and (3) have no issues as we are doing the same thing to both sides. There is no issue with line (4) as the difference of two squares factorisation (and the simple factorisation on the right hand side) hold for all a and b . The first problem occurs in line (5) as since we have assumed $a = b$, we have divided by $a - b$ which is 0. Therefore, the correct answer is **Option E**.

13.

If applying a function to both sides of an inequality, the inequality sign reverses if and only if the function is decreasing, so both Option A and Option B are sufficient and necessary. $\log_c x$ is only decreasing when $0 < c < 1$ so the answer is **Option C**.

14.

If he breaks his promise, he only needs to break one out of the three different STEP conditions. He does not need to have broken all three. Therefore, we can eliminate Option A and Option C as they contain the conjunction 'and' instead of 'or'. Option D is clearly incorrect as, for example, doing 4 STEP I questions breaks the promise. Therefore, **Option B** is correct. Note that Option B is necessary but not sufficient.

15.

The roots of the equation occur when $3 - 4 \sin x = 0 \Rightarrow \sin x = \frac{3}{4}$ which has two solutions greater than 0 and less than π (evident from the sin curve). Therefore, the answer is **Option C** as it is the only option with two roots in this range.

16.

If none of the Backstreet Boys want it that way, then all of them are lying, but zero of them wanting it that way is consistent with their lies, so this case is possible.

If one of the Backstreet Boys wants it that way, then one of them is telling the truth, which can only be Backstreet Boy A, which is consistent with the others' lies, so this case is possible.

If two or more of the Backstreet Boys want it that way, then multiple of them are telling the truth, which causes a contradiction as each specify an exact number of people who want it that way, so this case is not possible. Therefore, either no Backstreet Boys want it that way, or only Backstreet Boy A does, so the answer is **Option B**.

17.

Let $A_n = A_1 + (n-1)d$ and $B_n = B_1 r^{n-1}$. $A_1 + d = B_1 r^2$, $A_1 + 3d = B_1 r$. We multiply the first equation by 3 and subtract the second from this to get $2A_1 = B_1 r(3r - 1)$ which gives **Option A**.

18.

We are looking for the value of $0.3^{0.5^{0.7^{0.9^{1.1^{\dots}}}}}$. $1.1^{1.3^{\dots}}$ clearly diverges to infinity as each base is greater than 1, so now we wish to find $0.3^{0.5^{0.7^{0.9^{\infty}}}}$. As $0.9 < 1$, this is clearly $0.3^{0.5^{0.7^0}} = 0.3^{0.5^1} = \sqrt{0.3}$, which is **Option C**.

19.

We are given in the question that there is only one polynomial that satisfies the given conditions. The five equations show that $f(x) = 6 - x$ but this is not a degree 5 polynomial (where the coefficient of x^5 is 1. Therefore, we need to add a degree 4 polynomial whose value is 0 when $x = 1, 2, 3, 4$, i.e. $f(x) = 6 - x + k(x-1)(x-2)(x-3)(x-4)$ for some constant k . We are given that the coefficient of x^5 is 1, so $k = 1$ and plugging in $x = 6$ gives $f(6) = 120$ which is **Option F**.

20.

$2025 = a^2 - b^2 = (a + b)(a - b)$. The prime factorisation of 2025 is $3^4 5^2$ and since we can choose from 5 instances of the factor 3 and 3 instances of the factor 5 (including 3^0 and 5^0), the number of factors of 2025 is 15. We would have needed to consider negative factors, but we are given that $a - b > 0$ so it cannot be the case that $a + b$ and $a - b$ are both negative. We have $2025 = 45^2$ which gives $a + b = 45, a - b = 45$. By considering the division of 2025 by its factors, we have that if $a + b > 45$ then $a - b < 45$. Also, if $a + b < 45$ then $a - b > 45$, but it is necessarily true that $a + b > a - b$ so this leads to a contradiction. Therefore, after considering the $a + b = 45, a - b = 45$ case, we are left with 14 cases, which can be split into 7 symmetric pairs of cases, but one case in each pair leads to a contradiction, leaving 7 solutions, which is **Option F**. We would've needed to consider the special case $a = 45, b = 0$ but the question specifies positive integers.